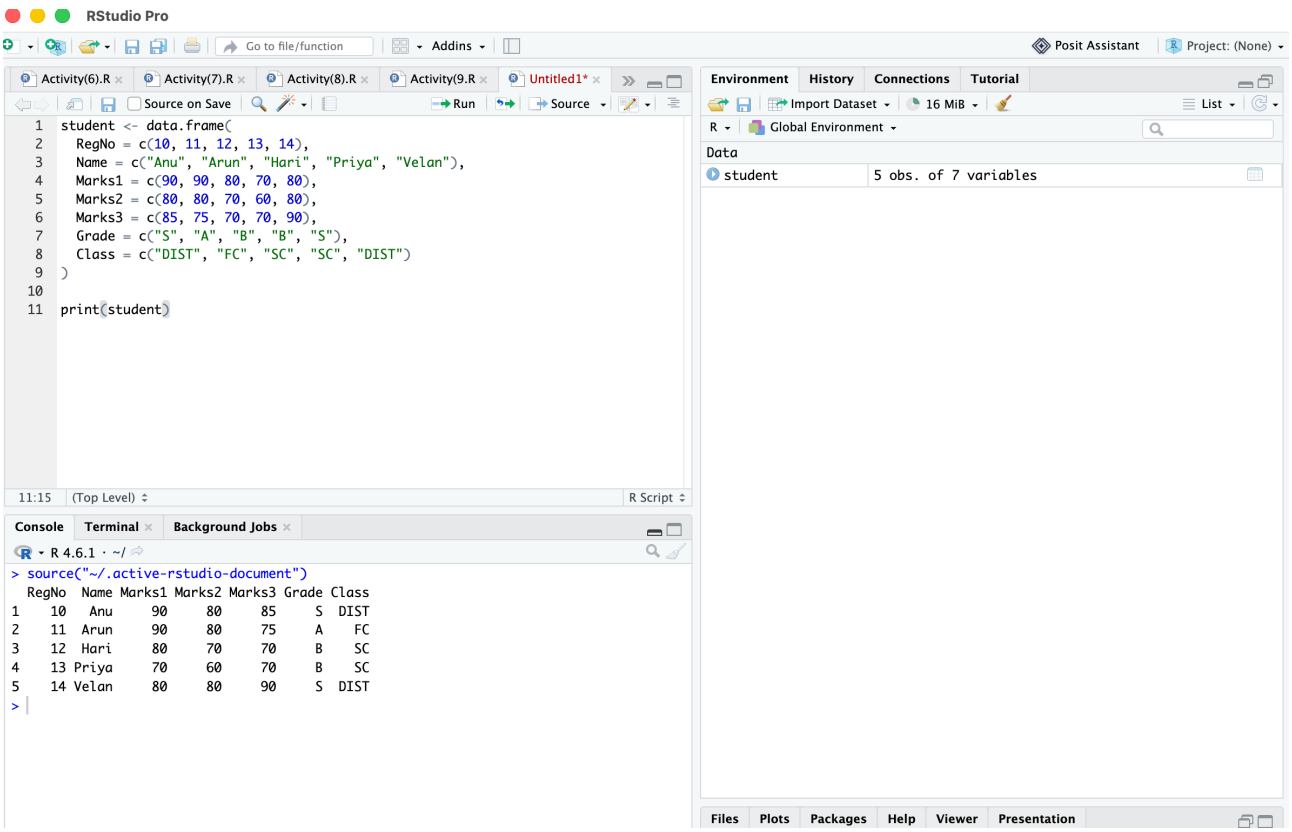
Name: N.R.Yasaswini

Register No: 192425119

1. Create a Data Frame

```
student <- data.frame(
  RegNo = c(10, 11, 12, 13, 14),
  Name = c("Anu", "Arun", "Hari", "Priya", "Velan"),
  Marks1 = c(90, 90, 80, 70, 80),
  Marks2 = c(80, 80, 70, 60, 80),
  Marks3 = c(85, 75, 70, 70, 90),
  Grade = c("S", "A", "B", "B", "S"),
  Class = c("DIST", "FC", "SC", "SC", "DIST")
)

print(student)
```



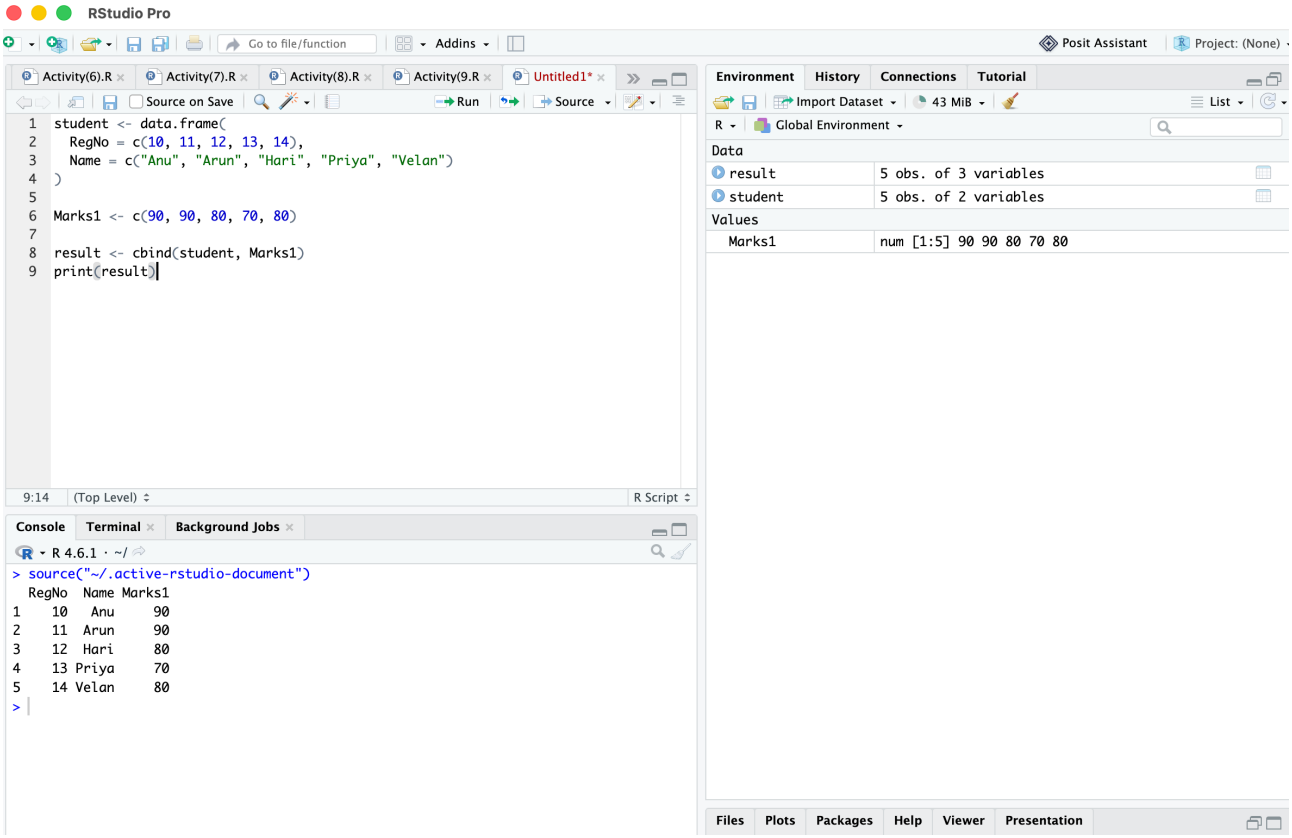
2. Using cbind()

```
student <- data.frame(
  RegNo = c(10, 11, 12, 13, 14),
  Name = c("Anu", "Arun", "Hari", "Priya", "Velan")
```

)

Marks1 <- c(90, 90, 80, 70, 80)

result <- cbind(student, Marks1)
print(result)
Explanation: cbind() adds columns to a data frame.

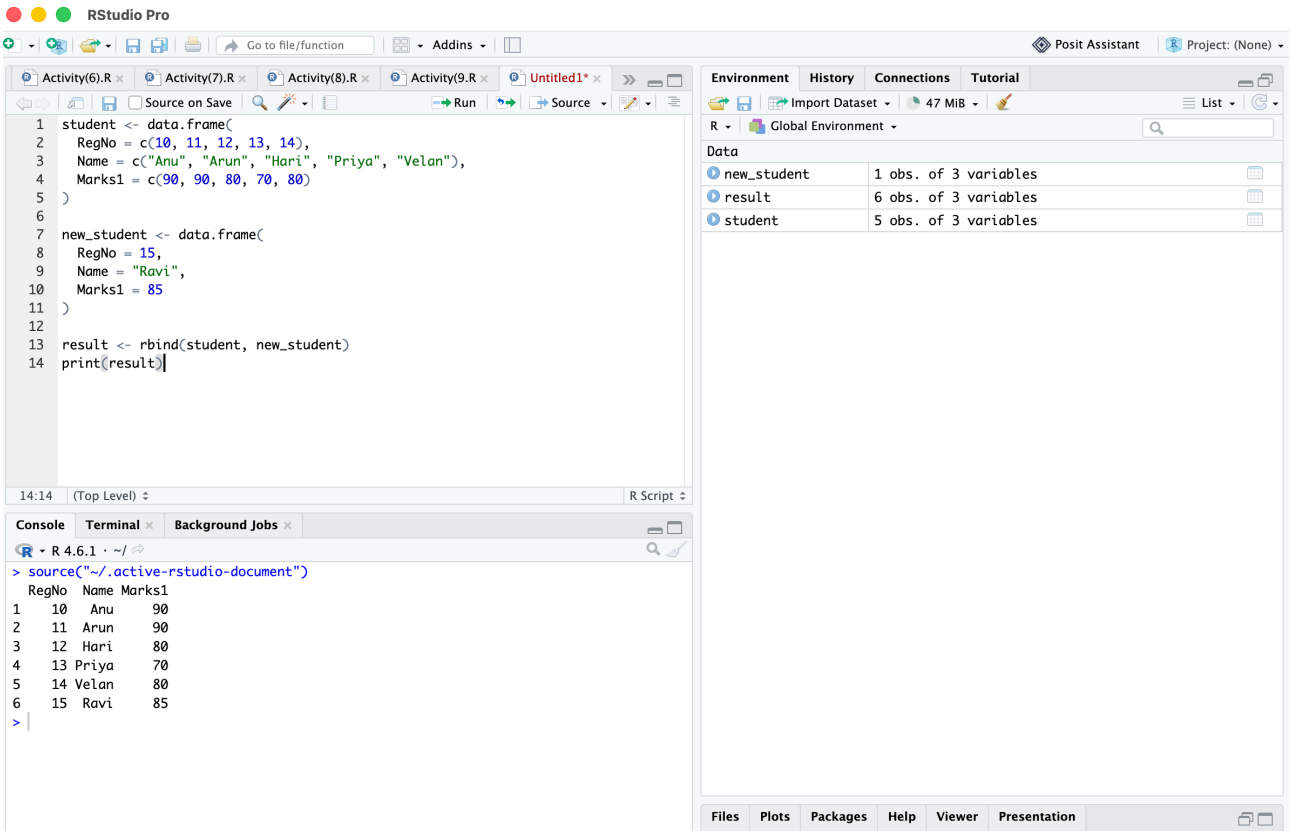


3. Using rbind()

student <- data.frame(
  RegNo = c(10, 11, 12, 13, 14),
  Name = c("Anu", "Arun", "Hari", "Priya", "Velan"),
  Marks1 = c(90, 90, 80, 70, 80)
)

new_student <- data.frame(
  RegNo = 15,
  Name = "Ravi",
  Marks1 = 85
)

result <- rbind(student, new_student)
print(result)
Explanation: rbind() adds a new row to a data frame.



4. Data Reshaping

Here, we use the built-in reshape() function to convert the marks columns into rows.

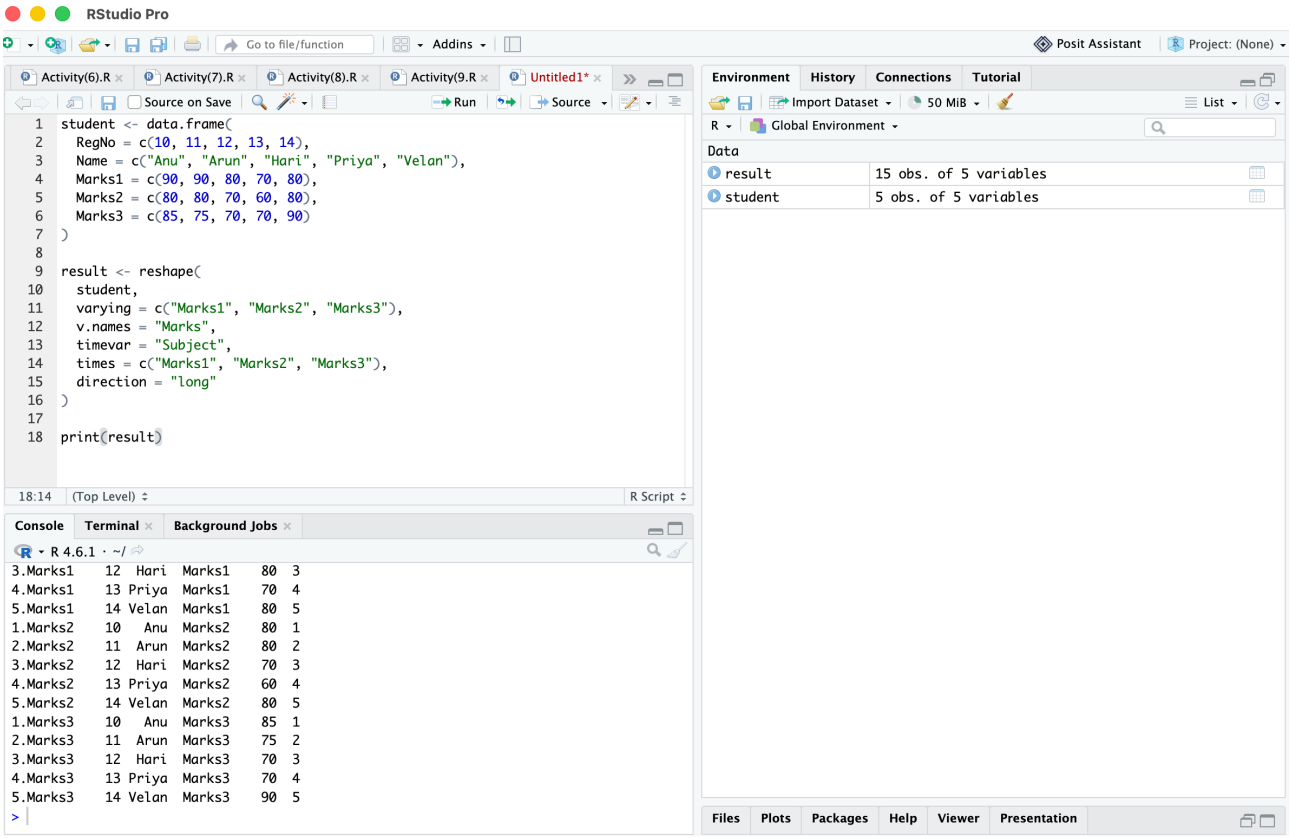
```
student <- data.frame(
  RegNo = c(10, 11, 12, 13, 14),
  Name = c("Anu", "Arun", "Hari", "Priya", "Velan"),
  Marks1 = c(90, 90, 80, 70, 80),
  Marks2 = c(80, 80, 70, 60, 80),
  Marks3 = c(85, 75, 70, 70, 90)
)

result <- reshape(
  student,
  varying = c("Marks1", "Marks2", "Marks3"),
  v.names = "Marks",
  timevar = "Subject",
  times = c("Marks1", "Marks2", "Marks3"),
```

```
  direction = "long"
)

print(result)
```

Explanation: reshape() converts wide-format data into long-format data.



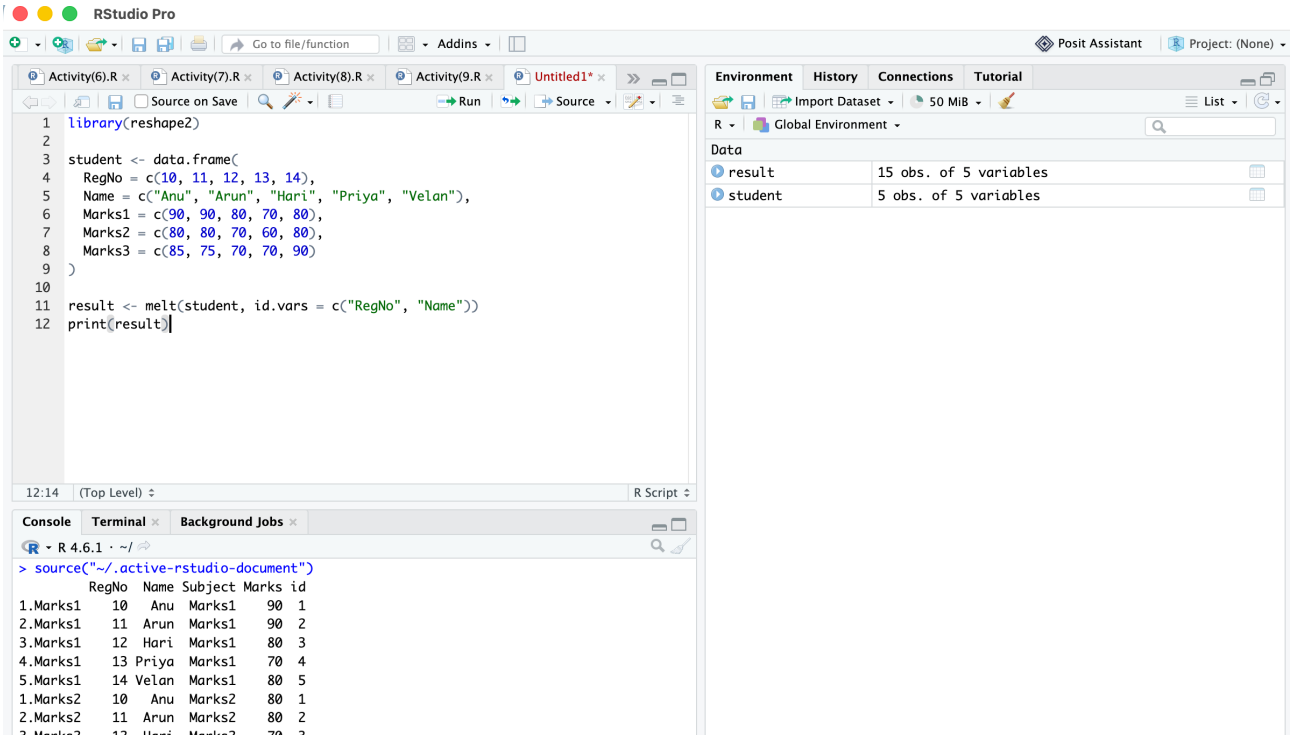
5. Using melt()

```
library(reshape2)

student <- data.frame(
  RegNo = c(10, 11, 12, 13, 14),
  Name = c("Anu", "Arun", "Hari", "Priya", "Velan"),
  Marks1 = c(90, 90, 80, 70, 80),
  Marks2 = c(80, 80, 70, 60, 80),
  Marks3 = c(85, 75, 70, 70, 90)
)

result <- melt(student, id.vars = c("RegNo", "Name"))
print(result)
```

Explanation: melt() converts the marks columns into rows and creates variable and value columns.



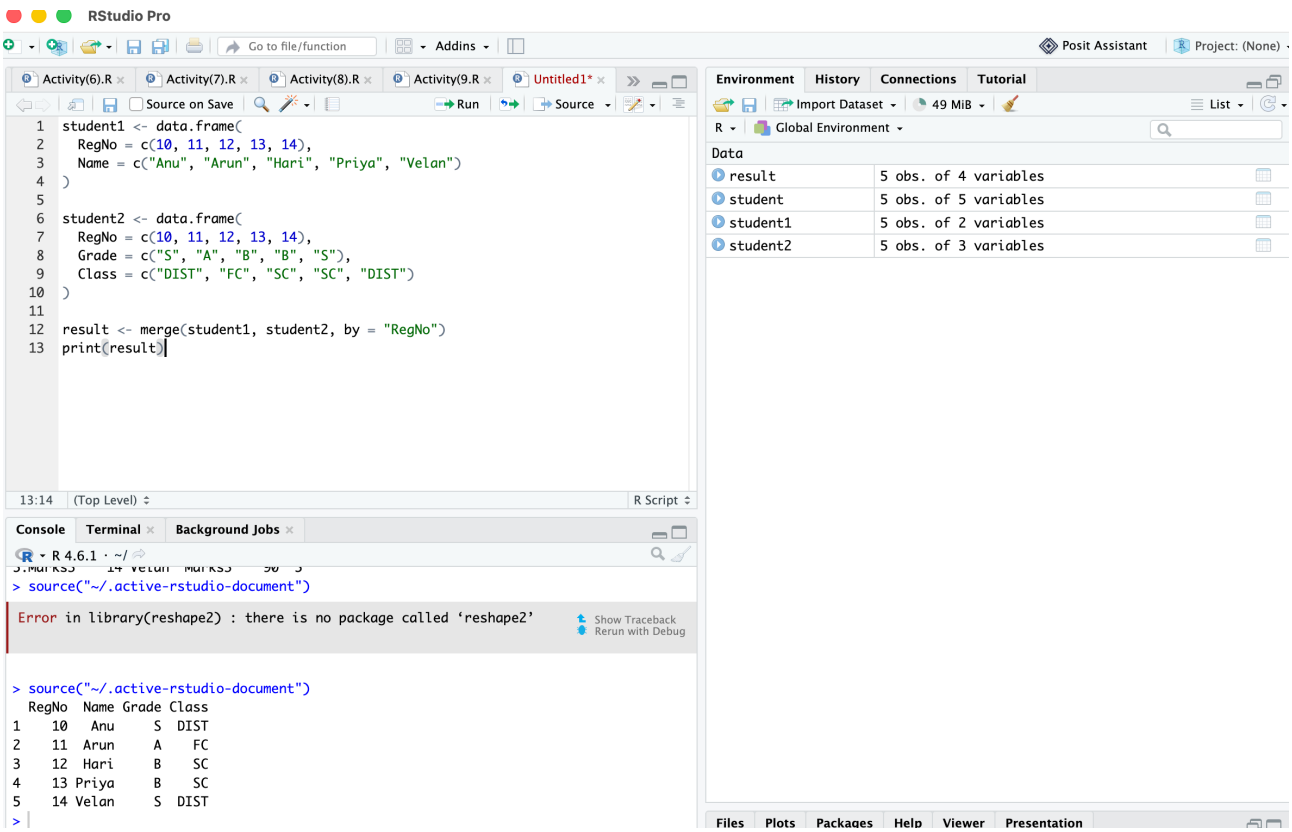
6. Using dcast()

```
library(reshape2)

student <- data.frame(
  RegNo = c(10, 11, 12, 13, 14),
  Name = c("Anu", "Arun", "Hari", "Priya", "Velan"),
  Marks1 = c(90, 90, 80, 70, 80),
  Marks2 = c(80, 80, 70, 60, 80),
  Marks3 = c(85, 75, 70, 70, 90)
)

long_data <- melt(student, id.vars = c("RegNo", "Name"))

result <- dcast(
  long_data,
  RegNo + Name ~ variable,
  value.var = "value"
)
```



print(result)
Explanation: dcast() converts long-format data back into wide-format data.

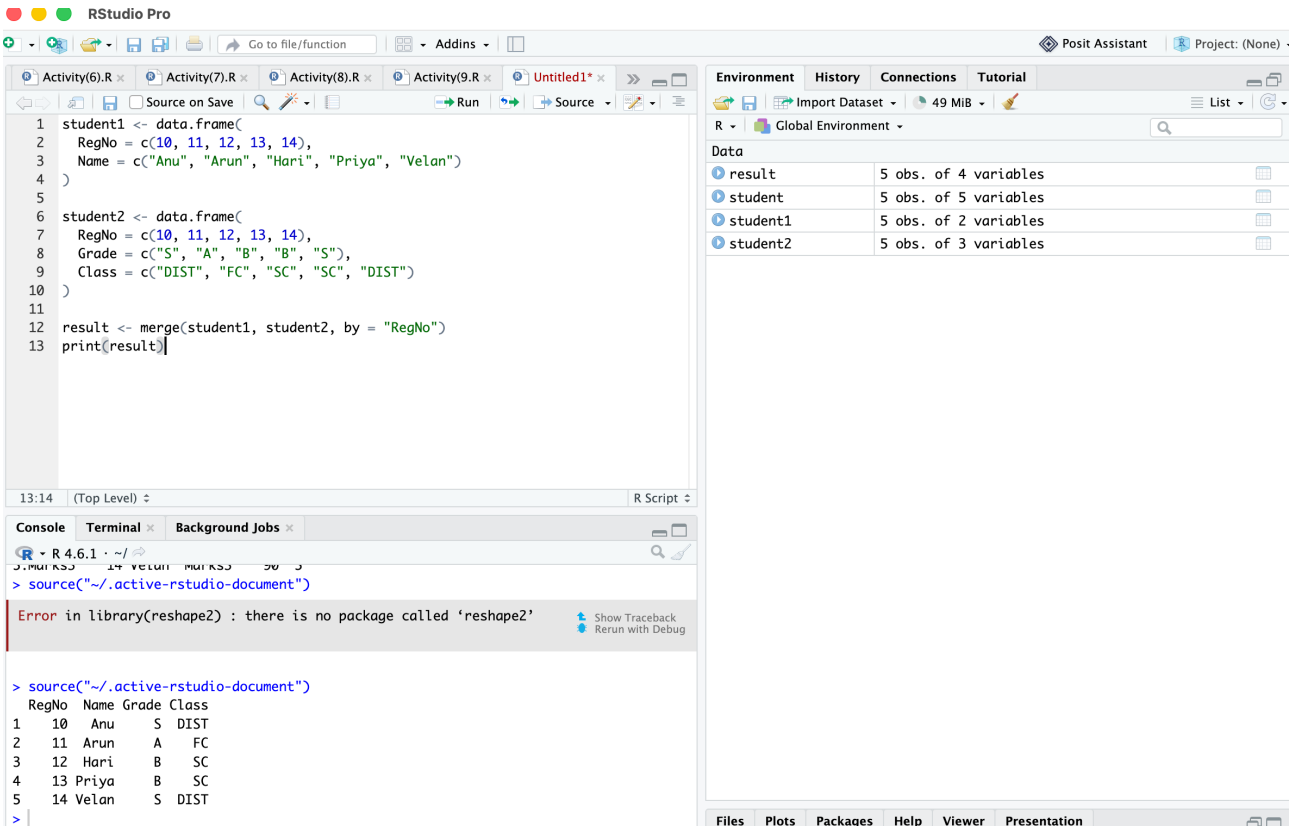
7. Using merge()

```
student1 <- data.frame(
  RegNo = c(10, 11, 12, 13, 14),
  Name = c("Anu", "Arun", "Hari", "Priya", "Velan")
)

student2 <- data.frame(
  RegNo = c(10, 11, 12, 13, 14),
  Grade = c("S", "A", "B", "B", "S"),
  Class = c("DIST", "FC", "SC", "SC", "DIST")
)

result <- merge(student1, student2, by = "RegNo")
print(result)
```

Explanation: merge() combines two data frames using the common column RegNo.


RStudio Pro
Activity(6).R
Activity(7).R
Activity(8).R
Activity(9.R
Untitled1*
Source on Save
Run
Source
1 student1 <- data.frame(
2 RegNo = c(10, 11, 12, 13, 14),
3 Name = c("Anu", "Arun", "Hari", "Priya", "Velan")
4)
5
6 student2 <- data.frame(
7 RegNo = c(10, 11, 12, 13, 14),
8 Grade = c("S", "A", "B", "B", "S"),
9 Class = c("DIST", "FC", "SC", "SC", "DIST")
10)
11
12 result <- merge(student1, student2, by = "RegNo")
13 print(result)
13:14
(Top Level)
R Script
Console
Terminal
Background Jobs
R 4.6.1 · ~/
> source("~/.active-rstudio-document")
Error in library(reshape2) : there is no package called 'reshape2'
Show Traceback
Rerun with Debug
> source("~/.active-rstudio-document")
RegNo Name Grade Class
1 10 Anu S DIST
2 11 Arun A FC
3 12 Hari B SC
4 13 Priya B SC
5 14 Velan S DIST
Environment
History
Connections
Tutorial
Import Dataset
49 MiB
List
R
Global Environment
Data
result 5 obs. of 4 variables
student 5 obs. of 5 variables
student1 5 obs. of 2 variables
student2 5 obs. of 3 variables
Files
Plots
Packages
Help
Viewer
Presentation